Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

Practical 3

About this unit

Practical 3

Practice Lab Assignment

Unit • 100% completed

Lab Assignment

Unit • 100% completed

Hot days ahead 27°C

Search

ENG IN

08:58 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d66/6684d40e5d95376955a73c1d

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.1.1. Numpy Array Operations

08:17

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases +

Explorer

numpyarr...

Submit

```
1  import numpy as np
2
3  # write your code here...
4  rows, cols=map(int,input().split())
5  elements=[]
6  for _ in range(rows):
7      elements.extend(map(int,input().split()))
8  array=np.array(elements).reshape(rows,cols)
9  print(array)
10 print(array.ndim)
11 print(array.shape)
12 print(array.size)
```

Debugger

Average time **0.012 s** 12.00 ms

Maximum time **0.021 s** 21.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 21 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]

Terminal Test cases

‹ Prev Reset Submit Next ›

Hot days ahead 27°C

Search

ENG IN 08:58 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774e3fff4ab787eca9da092

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.2.1. Numpy: Matrix Operations

09:11

The given code takes two 3×3 matrices, $\mathrm{matrix\_a}$, and $\mathrm{matrix\_b}$, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ($\mathrm{matrix\_a} + \mathrm{matrix\_b}$)
2. **Subtraction** ($\mathrm{matrix\_a} - \mathrm{matrix\_b}$)
3. **Element-wise Multiplication** ($\mathrm{matrix\_a} * \mathrm{matrix\_b}$)
4. **Matrix Multiplication** ($\mathrm{matrix\_a} \cdot \mathrm{matrix\_b}$)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for $\mathrm{matrix\_a}$, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for $\mathrm{matrix\_b}$, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Explorer

matrixOp...

Submit

Debugger

```
1  import numpy as np
2
3  # Input matrices
4  print("Enter Matrix A:")
5  matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
6
7  print("Enter Matrix B:")
8  matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
9
10
11 # Addition
12 sum=matrix_a+matrix_b
13 print("Addition (A + B):")
14 print(sum)
15
```

Average time: 0.024 s, 24.25 ms

Maximum time: 0.028 s, 28.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 28 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3

Terminal Test cases

‹ Prev Reset Submit Next ›

Hot days ahead 27°C

Search

ENG IN

08:59 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774ee19f4ab787eca9da7fe

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

11:49

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases +

Explorer

stacking.py

Submit

```
1  import numpy as np
2
3  # Input matrices
4  print("Enter Array1:")
5  arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7  print("Enter Array2:")
8  arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 horizontal_stack=np.hstack((arr1,arr2))
13 print(horizontal_stack)
14 # Perform vertical stacking (vstack)
15 print("Vertical Stack:")
16 vertical_stack=np.vstack((arr1,arr2))
17 print(vertical_stack)
```

Debugger

Average time 0.024 s 24.25 ms

Maximum time 0.028 s 28.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 28 ms Debug

Expected output Actual output

Terminal Test cases

‹ Prev Reset Submit Next ›

Hot days ahead 27°C

Search

ENG IN 08:59 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?euclId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774ee9df4ab787eca9da869

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.2.3. Numpy: Custom Sequence Generation

04:54

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases +

Explorer

customS...

Submit

```
1  import numpy as np
2
3  # Take user input for the start, stop, and step of the sequence
4  start = int(input())
5  stop = int(input())
6  step = int(input())
7
8  # Generate the sequence using np.arange()
9  sequence=np.arange(start,stop,step)
10 # Print the generated sequence
11 print(sequence)
```

Debugger

Average time **0.014 s** 14.25 ms

Maximum time **0.018 s** 18.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 18 ms Debug

Expected output	Actual output
3	3
10	10
2	2
[3 5 7 9]	[3 5 7 9]

Terminal Test cases

‹ Prev Reset Submit Next ›

Hot days ahead 27°C

Search

ENG IN 08:59 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774ef2cf4ab787eca9da8c0

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitw... 11:57

You are given two arrays `A` and `B`. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. **Arithmetic Operations**:
   - Compute the element-wise sum, difference, and product of the two arrays.
2. **Statistical Operations**:
   - Calculate the mean, median, and standard deviation of array `A`.
3. **Bitwise Operations**:
   - Perform bitwise `AND`, bitwise `OR`, and bitwise `XOR` on the arrays (ex: A_i `OR` B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array `A`.
- The second line contains space-separated integers representing the elements of array `B`.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases +

Explorer

different...

Submit

Debugger

```
3  def array_operations(A, B):
4
5      # Convert A and B to NumPy arrays
6      A=np.array(A)
7      B=np.array(B)
8      # Arithmetic Operations
9      sum_result = A+B
10     diff_result = A-B
11     prod_result = A*B
12
13     # Statistical Operations
14     mean_A = np.mean(A)
15     median_A = np.median(A)
16     std_dev_A = np.std(A)
17
18     # Bitwise Operations
19     and_result = A & B
20     or_result = A | B
21     xor_result = A ^ B
22
```

Average time **0.020 s** 20.00 ms

Maximum time **0.028 s** 28.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Terminal Test cases

‹ Prev Reset Submit Next ›

Hot days ahead 27°C

Search

ENG IN 09:00 07-05-2026

3.2.5. Numpy: Copying and Viewing Arrays

05:36

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: <original_array>
View array: <view_array>
```

- After modifying the copy:

```
Original array after modifying copy: <original_array>
Copy array: <copy_array>
```

Sample Test Cases +

Explorer

copyAnd...

```
1  import numpy as np
2
3  inputlist = list(map(int,input().split(" ")))
4
5  # Original array
6  original_array = np.array(inputlist)
7
8  # Create a view
9  view_array =original_array.view()
10
11 # Create a copy
12 copy_array =original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_
17 print("View array:", view_array)
```

Average time 0.007 s 7.25 ms | Maximum time 0.010 s 10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output Actual output

Terminal Test cases

Prev

Hot days ahead 27°C Search

Course

Course
mitaoe.codetantra.com

Memory usage: 379 MB

Ask Gemini

a.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774efaaf4ab787eca9da91e

School

Learn Anywhere

202501090065@mitaoe.ac.in Support Logout

3.2.5. Numpy: Copying and Viewing Arrays

05:36

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: <original_array>
View array: <view_array>
```

- After modifying the copy:

```
Original array after modifying copy: <original_array>
Copy array: <copy_array>
```

Sample Test Cases +

Explorer

copyAnd...

Submit

```
1  import numpy as np
2
3  inputlist = list(map(int,input().split(" ")))
4
5  # Original array
6  original_array = np.array(inputlist)
7
8  # Create a view
9  view_array =original_array.view()
10
11 # Create a copy
12 copy_array =original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_array)
17 print("View array:", view_array)
```

Debugger

Average time **0.007 s** 7.25 ms

Maximum time **0.010 s** 10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output Actual output

Terminal Test cases

‹ Prev Reset Submit Next ›

27°C Sunny

Search

ENG IN

09:01 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774f064f4ab787eca9da9b0

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

07:47

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where `search_value` appears in `array1` and print these indices.
2. Counting: Count how many times `count_value` appears in `array1` and print the count.
3. Broadcasting: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
4. Sorting: Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.

Explorer

arrayOpe...

Submit

Debugger

```
1  import numpy as np
2
3  # Input array from the user
4  array1 = np.array(list(map(int, input().split())))
5
6  # Searching
7  search_value = int(input("Value to search: "))
8  count_value = int(input("Value to count: "))
9  broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 A=np.where(array1==search_value)[0]
13 print(A)
14 # Count occurrences in array1
15 B=np.count_nonzero(array1==count_value)
16 print(B)
17 # Broadcasting addition
18 C= array1+broadcast_value
19 print(C)
20 # Sort the first array
```

Average time **0.022 s** 22.50 ms

Maximum time **0.029 s** 29.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Terminal Test cases

‹ Prev Reset Submit Next ›

27°C Sunny

Search

ENG IN

09:01 07-05-2026

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/67dfb4f1d1a74a4035ef4225

School

CODETANTRA Home Learn Anywhere 202501090065@mitaoe.ac.in Support Logout

3.2.7. Student Data Analysis and Operations

28:49

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of

Explorer

Operatio...

Submit

Debugger

```python
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
# 1. Print all student details
print("All student Details:\n", a)

# 2. Print total students
print("Total Students:", len(a))

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:, 0])

# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])

# 5. Print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6. Print maximum marks of Subject 3
```

Average time 0.025 s 25.00 ms

Maximum time 0.025 s 25.00 ms

1 out of 1 shown test case(s) passed

Test case 1 25 ms Debug

Terminal Test cases

‹ Prev Reset Submit Next ›

27°C Sunny

Search

ENG IN

09:02 07-05-2026